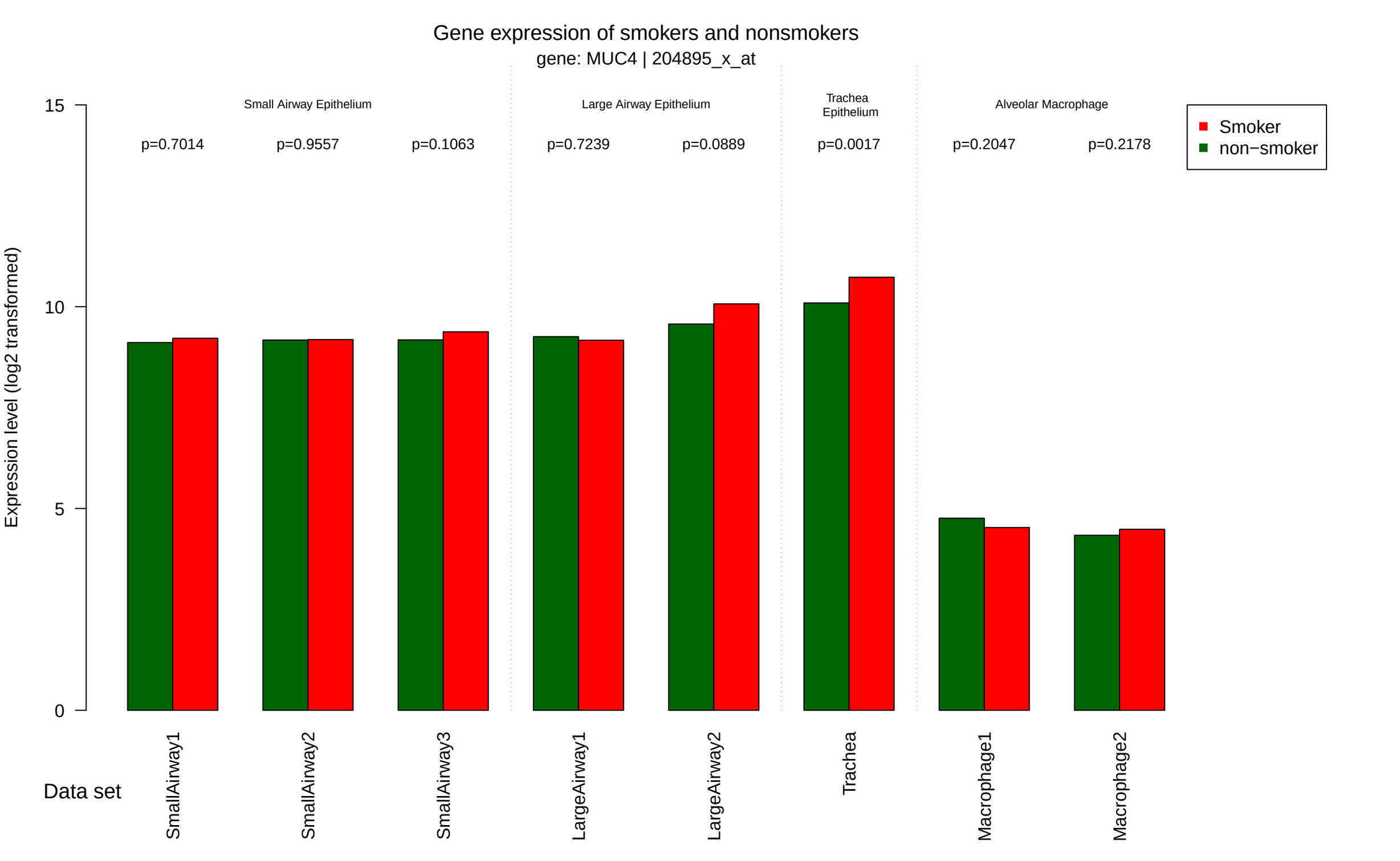
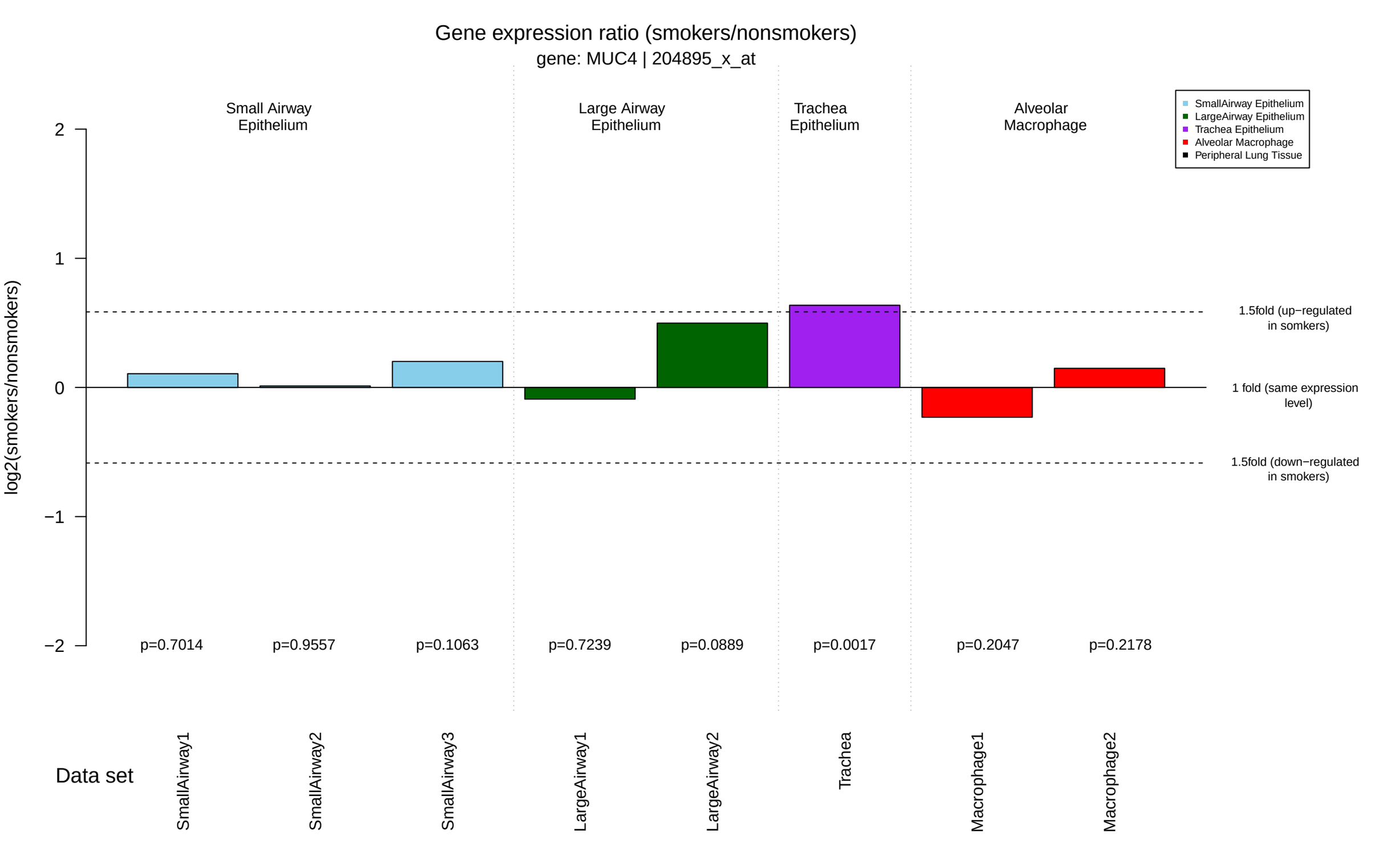
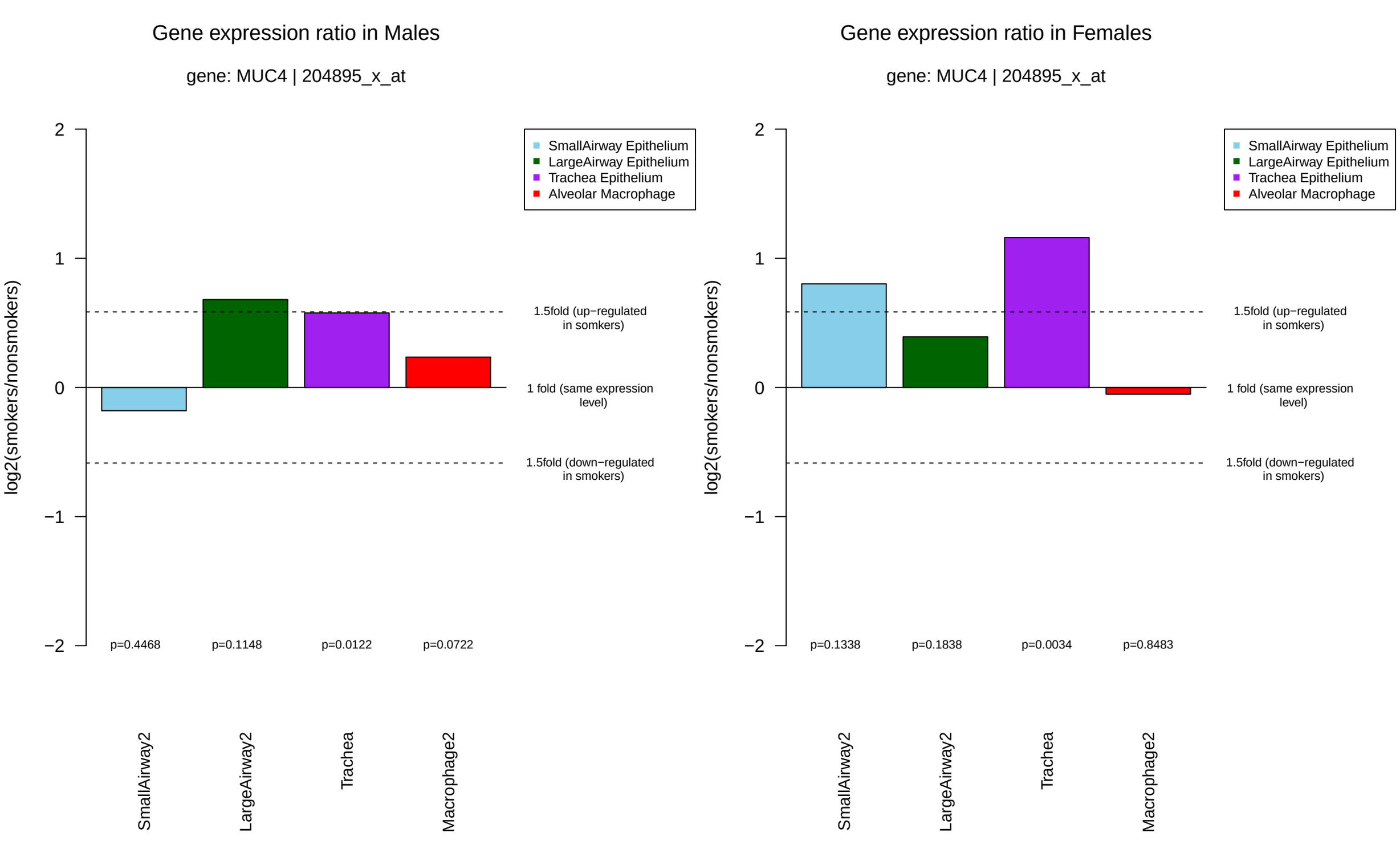
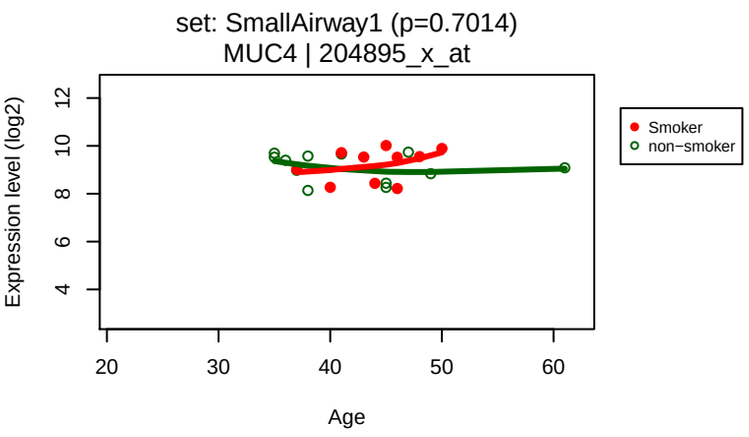
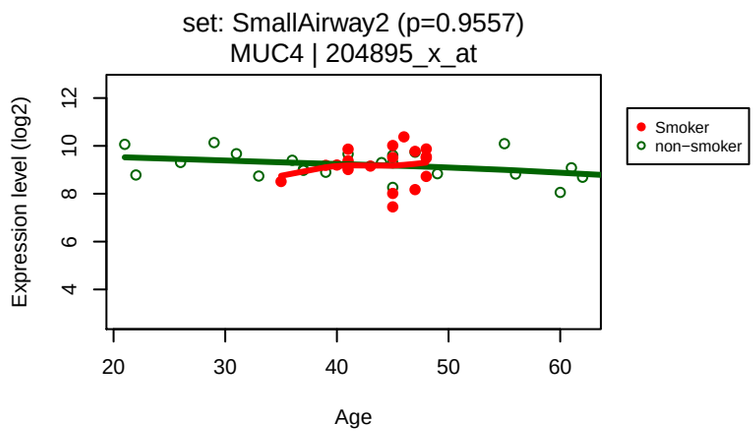
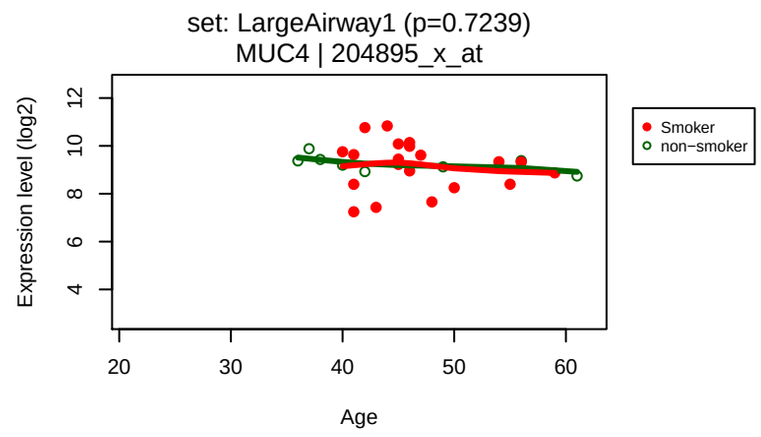

Gene expression of smokers and nonsmokers
gene: MUC4 | 204895_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7014
p=0.9557
p=0.1063
p=0.7239
p=0.0889
p=0.0017
p=0.2047
p=0.2178
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

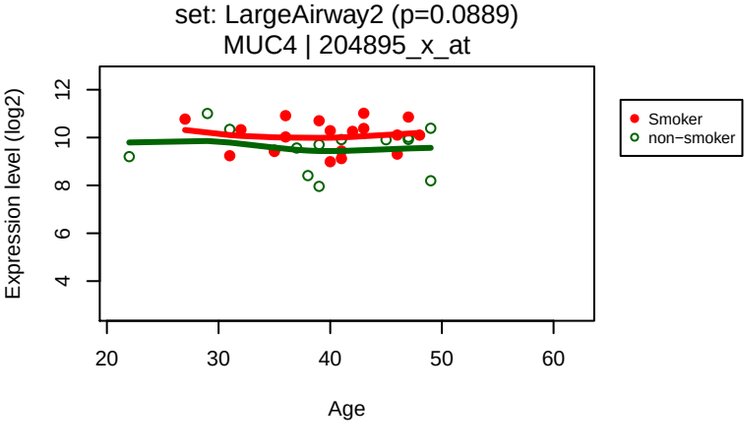

Gene expression ratio (smokers/nonsmokers)
gene: MUC4 | 204895_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7014
p=0.9557
p=0.1063
p=0.7239
p=0.0889
p=0.0017
p=0.2047
p=0.2178
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

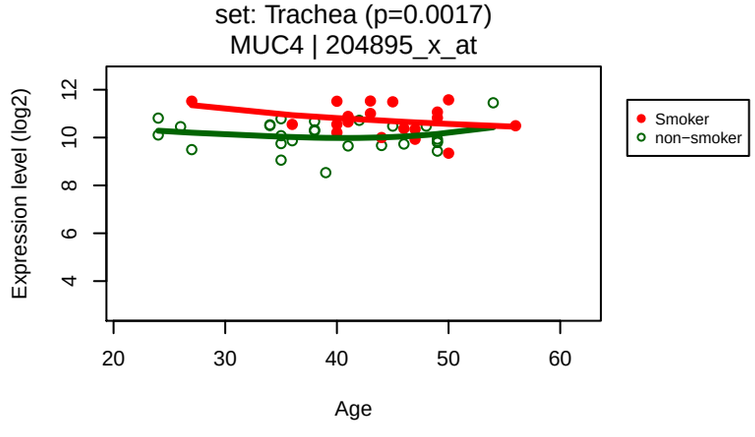

Gene expression ratio in Males
gene: MUC4 | 204895_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4468
p=0.1148
p=0.0122
p=0.0722
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MUC4 | 204895_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1338
p=0.1838
p=0.0034
p=0.8483
SmallAirway2
LargeAirway2
Trachea
Macrophage2

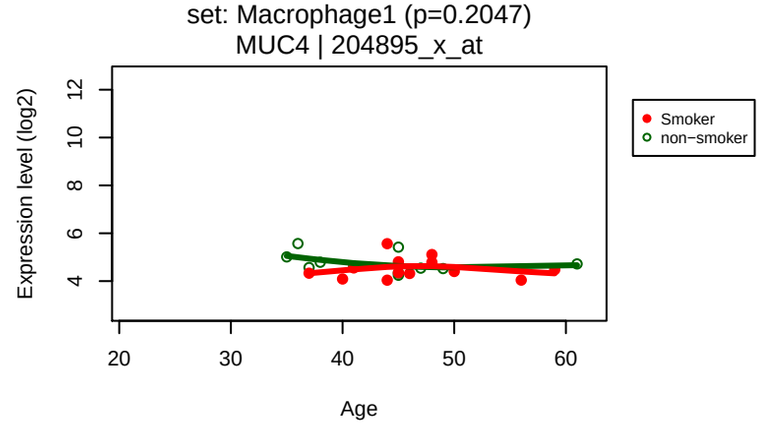

set: SmallAirway1 (p=0.7014)
MUC4 | 204895_x_at
Expression level (log2)
Age
Smoker
non−smoker

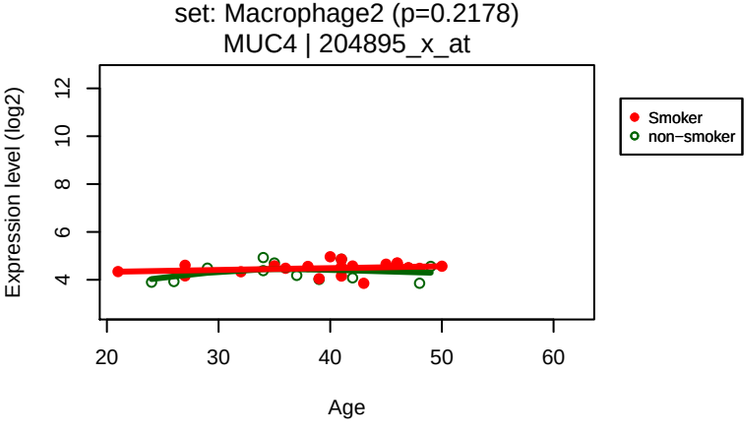

set: SmallAirway2 (p=0.9557)
MUC4 | 204895_x_at
Expression level (log2)
Age
Smoker
non−smoker

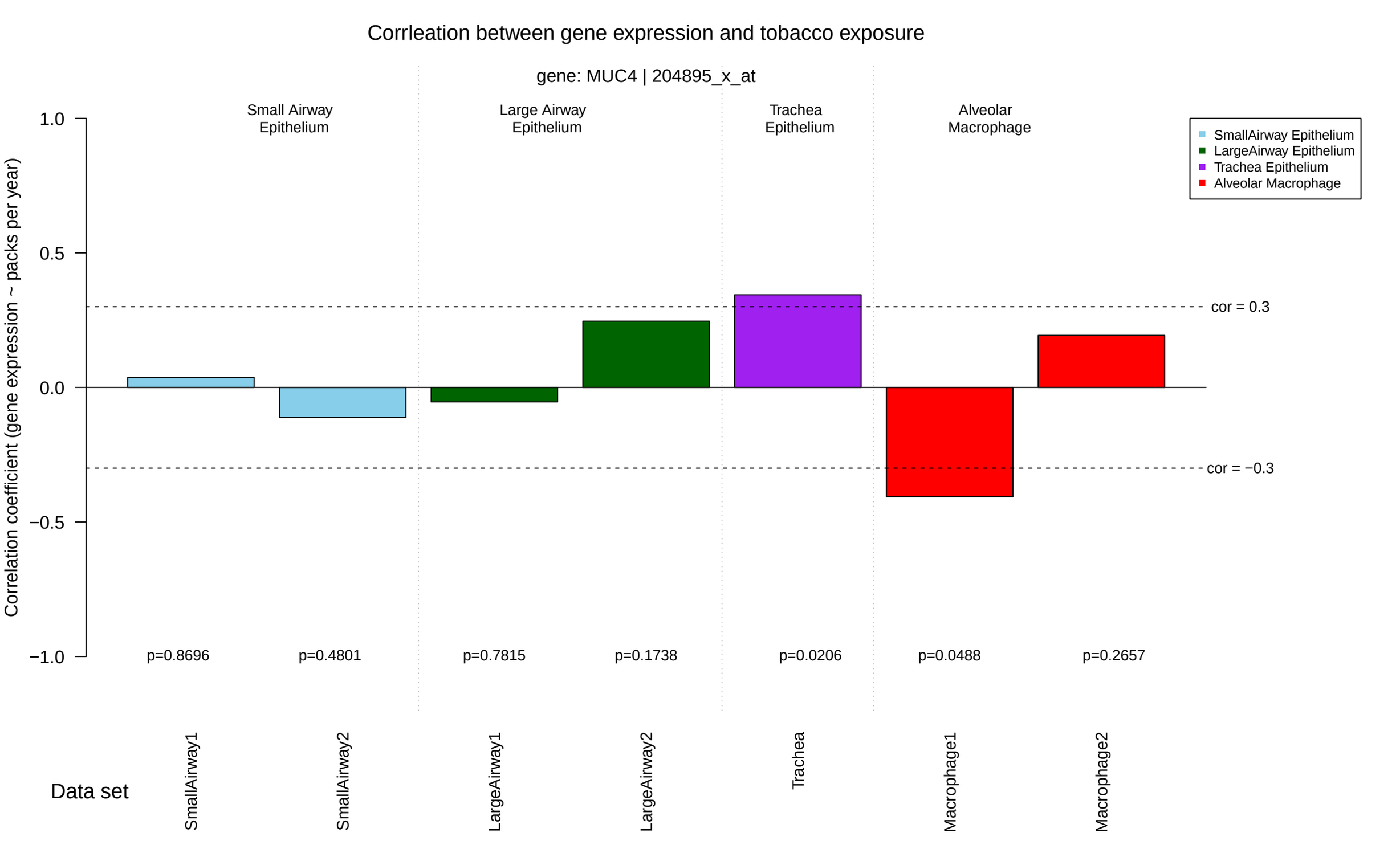

set: LargeAirway1 (p=0.7239)
MUC4 | 204895_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0889)
MUC4 | 204895_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0017)
MUC4 | 204895_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2047)
MUC4 | 204895_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2178)
MUC4 | 204895_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MUC4 | 204895_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8696
p=0.4801
p=0.7815
p=0.1738
p=0.0206
p=0.0488
p=0.2657
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2